



ROBSTRIDE DYNAMICS

MOTOR INNOVATION LEADS DYNAMIC EVOLUTION

一体化青春版高性能关节模组

北京灵足时代科技有限公司
RobStride Dynamics Co., Ltd

邮箱: shaoyuanxin@robstride.com
官网: www.robstride.com

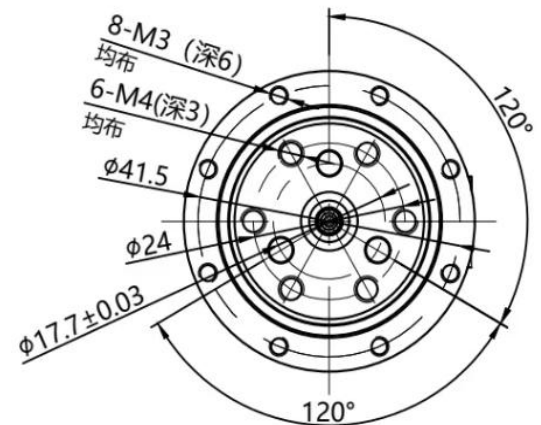
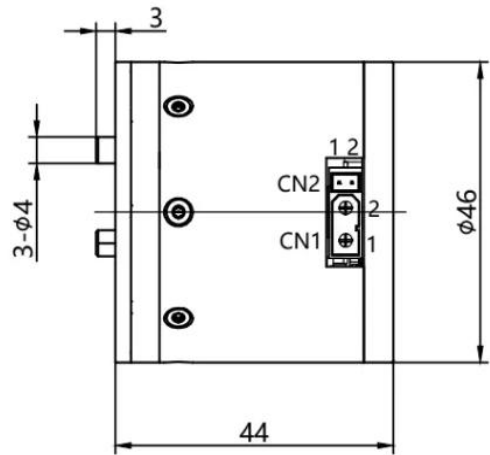
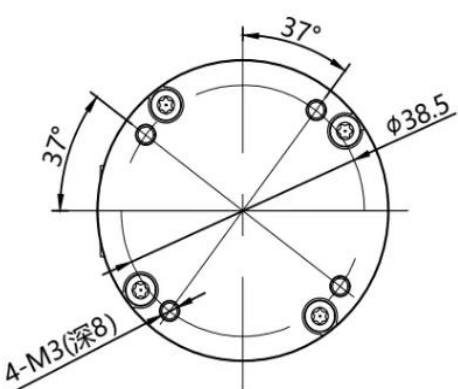


EDULITE 05 - ¥ 299

准直驱6N.m一体化电机模组
6N.m Integrated QDD Actuator



尺寸参数 Dimension s



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机械参数 Mechanical Specs

科目 Item	参数 Spec
重量 Weight	242g
极数 Pole	20
相数 Phase	3
驱动方式 Drive Method	FOC
减速比 reduction ratio	9:1
齿轮工艺 Gear Material	粉末冶金 Powder Metallurgy



电气特性 Electrical Specs

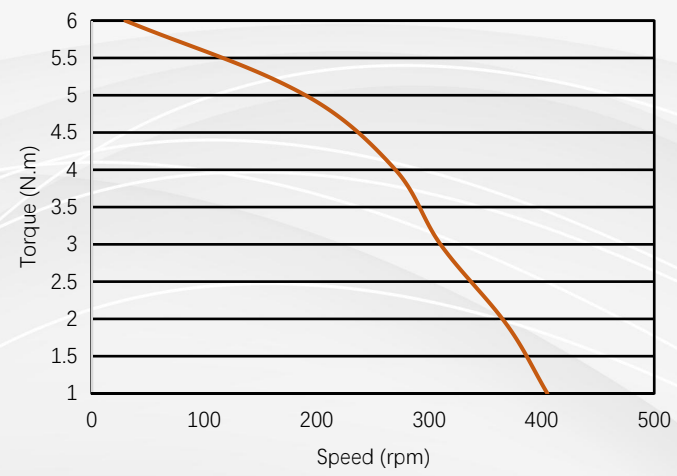
科目 Item	参数 Spec
磁编码器 Magnetic Encoder	2pcs
额定电压 Rated Voltage	48V
使用电压范围 Voltage Range	15-60V
额定输出功率 Rated Output Power	19W±10%
额定扭矩 Rated Torque (100rpm下测得)	1.9N.m (150mm×150mm铝散热板)
峰值扭矩 Peak Torque	6N.m
运转方向 Running Direction	CW/CCW
使用温度范围 Temperature Range	-20~50℃
使用湿度范围 Humidity Range	5~85%
保存温度范围 Storage Temperature Range	-30~70℃
绝缘等级 Insulation Level	Class B
空载转速 No-load Speed	430 rpm ± 10%
空载电流 No-load Current	0.14 Arms ±10%
峰值相电流 Peak Phase Current	11Apk ± 10%
绝缘电阻/定子绕组 Insulation Resistance/ Stator Winding	DC 500VAC, 100M Ohms
耐高压/定子与机壳 High Voltage Resistance/ Stator and Casing	600VAC, 1s, 2mA
电机反电势 Back-EMF	7.4Vrms/krpm ±10%
线电阻 Line Resistance	2.72Ω ± 10%
转矩常数 Torque Constant	0.94N.m/Arms
电机电感 Inductance	0.813mH ±10%
低速端等效惯量 Moment Inertia	9.4e-4kg*m2



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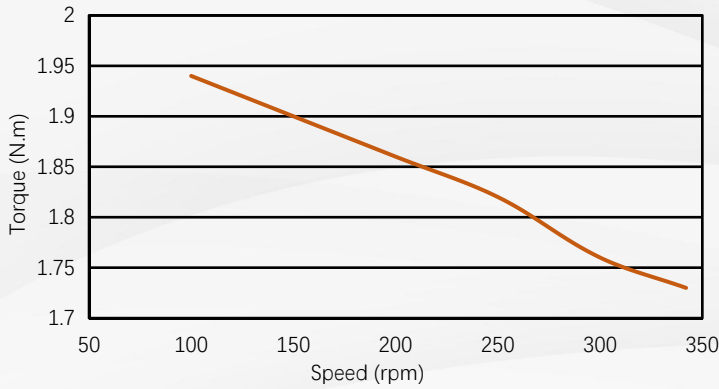
性能曲线
Performance Curves

48V T-N曲线 (T-N under 48V)



旋转负载 Load (N.m)	转速 Rotational Speed(rpm)
1	405
1.9	370
3	310
4	270
5	190
6	30

热平衡态T-N曲线
Thermal equilibrium state T-N curve
(150mm×150mm铝散热板)



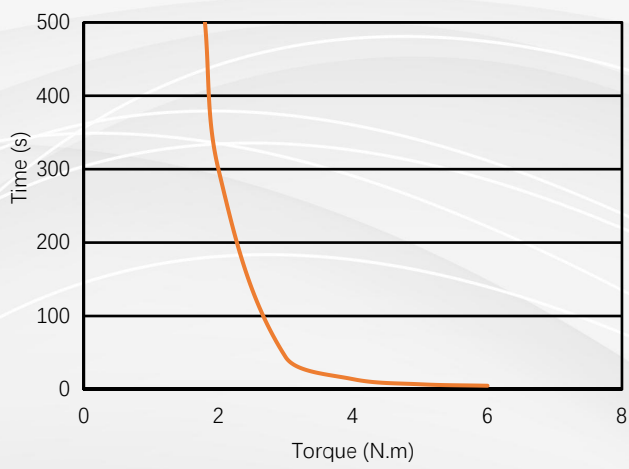
力矩 Torque(N.m)	转速 Rotational Speed(rpm)
1.73	342
1.76	300
1.82	250
1.86	200
1.9	150
1.94	100



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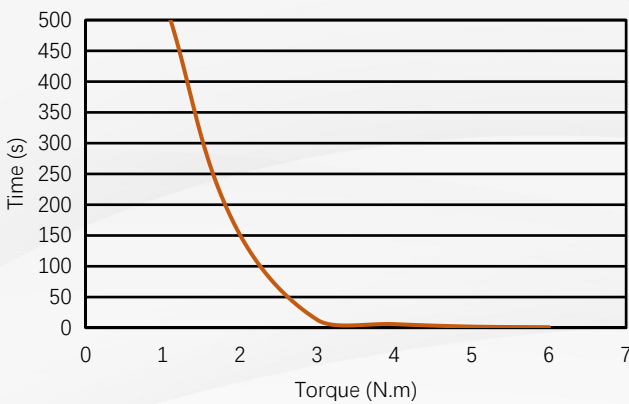
性能曲线
Performance Curves

旋转工况最大过载曲线 (Max Overload)
(150mm×150mm铝散热板)



旋转负载 Load (N.m)	工作时间 Operating Time (s)
1.9	rated
2	300
3	44
4	14
5	7
6	5

堵转工况最大过载曲线 (Max Overload)



堵转负载 Load (N.m)	工作时间 Operating Time (s)
1.1	rated
1.9	175
3	13
4	6
5	2
6	1